



Data, Models, and Magic: Getting Started with Power BI

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BI Architect @ **Lucient**

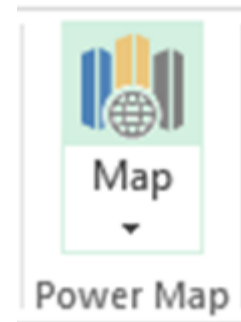
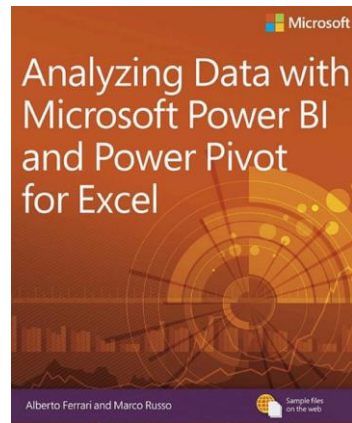
+15 years in Microsoft “Data Realm”

Giving my little contribution to the community





A little bit of history



2010

2012

2013

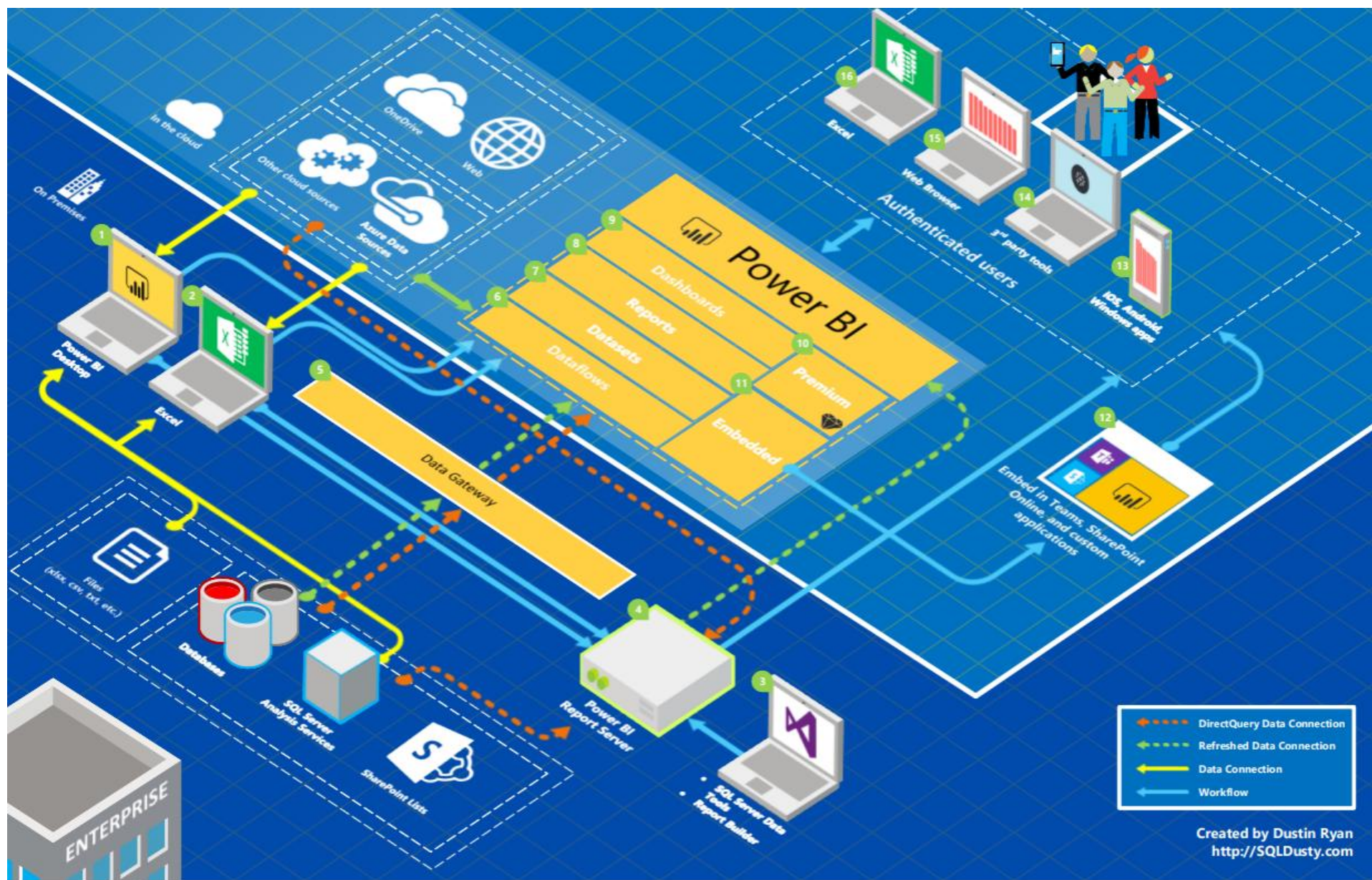
2015



Power BI is not your toy boy



Power BI is not a tool
Power BI is an ecosystem



Power BI Architecture Diagram v4 is now available! | Data and Analytics with Dustin Ryan

THE ICEBERG OF ASSUMPTIONS WITH POWER BI



[Does it feel like too much? — DATA GOBLINS](#)



It is low code/no code,
isn't it?

You'll write code if you need it

- ETL → Power Query / M
- Modeling → DAX / TMSL / TMDL / C#
- Reporting → Basic Fx Support
- Paginated Reporting → VBScript



Power Query / M

New ETL approach

Power Query

Home Transform Add column View Help

Get data Enter data Manage connections Options Manage parameters Refresh Advanced editor Choose columns Remove columns Keep rows Remove rows Filter rows Sort Suggested transforms Split column Group by Data type: Text Use first row as headers Replace values Combine Map to entity AI insights Export template

Queries [1]

Customers

Source[Name = "Customers", Signature = "table"] [Data]

	CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	PostalCode
1	ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Obere Str. 57	Berlin	null	12209
2	ANATR	Ana Trujillo Emparedados y helados	Ana Trujillo	Owner	Avda. de la Constitución 2222	México D.F.	null	05021
3	ANTON	Antonio Moreno Taquería	Antonio Moreno	Owner	Mataderos 2312	México D.F.	null	05023
4	AROUT	Around the Horn	Thomas Hardy	Sales Representative	120 Hanover Sq.	London	null	WA1 1DP
5	BERGS	Berglunds snabbköp	Christina Berglund	Order Administrator	Berguvsvägen 8	Luleå	null	S-958 22
6	BLAUS	Blauer See Delikatessen	Hanna Moos	Sales Representative	Forsterstr. 57	Mannheim	null	68306
7	BLONP	Blondesd's père et fils	Frédérique Citeaux	Marketing Manager	24, place Kléber	Strasbourg	null	67000
8	BOLID	Bólido Comidas preparadas	Martin Sommer	Owner	C/ Araquil, 67	Madrid	null	28023
9	BONAP	Bon app'	Laurence Leblan	Owner	12, rue des Bouchers	Marseille	null	13008
10	BOTTM	Bottom-Dollar Markets	Elizabeth Lincoln	Accounting Manager	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4
11	BSBEV	B's Beverages	Victoria Ashworth	Sales Representative	Fauntleroy Circus	London	null	EC2 5NT
12	CACTU	Cactus Comidas para llevar	Patricio Simpson	Sales Agent	Cerrito 333	Buenos Aires	null	1010
13	CENTC	Centro comercial Moctezuma	Francisco Chang	Marketing Manager	Sierras de Granada 9993	México D.F.	null	05022
14	CHOPS	Chop-suey Chinese	Yang Wang	Owner	Hauptstr. 29	Bern	null	3012
15	COMMI	Comércio Mineiro	Pedro Afonso	Sales Associate	Av. dos Lusíadas, 23	Sao Paulo	SP	05432-043
16	CONSH	Consolidated Holdings	Elizabeth Brown	Sales Representative	Berkeley Gardens 12 Brewery	London	null	WX1 6LT
17	DRACD	Drachenblut Delikatessen	Sven Ottlieb	Order Administrator	Walsenweg 21	Aachen	null	52066
18	DUMON	Du monde entier	Janine Labrun	Owner	67, rue des Cinquante Otages	Nantes	null	44000
19	EASTC	Eastern Connection	Ann Devon	Sales Agent	35 King George	London	null	WX3 6FW
20	ERNSH	Ernst Handel	Roland Mendel	Sales Manager	Kirchgasse 6	Graz	null	8010
21	FAMIA	Familia Arquibaldo	Aria Cruz	Marketing Assistant	Rua Orós, 92	Sao Paulo	SP	05442-030
22	FISSA	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel	Accounting Manager	C/ Moralzarzal, 86	Madrid	null	28034
23	FOLIG	Folies gourmandes	Martine Rancé	Assistant Sales Agent	184, chaussée de Tournai	Lille	null	59000
24	FOLKO	Folk och få HB	Maria Larsson	Owner	Åkergatan 24	Bräcke	null	S-844 67
25	FRANM	Francher	Martin Sommer	Marketing Manager	Sierras de Granada 9993	México D.F.	null	05022

Query settings

Properties

Name: Customers

Entity type: Custom

Applied steps

Source

Navigation 1

1 warning Columns: 13 Rows: 91

Cancel Save & close

1. Ribbon
2. Queries pane
3. Current data status
4. Query settings
5. Status bar

Behind the scenes

Advanced editor



```
1 let Orders = Table.FromRecords({  
2   [OrderID = 1, CustomerID = 1, Item = "fishing rod", Price = 100.0],  
3   [OrderID = 2, CustomerID = 1, Item = "1 lb. worms", Price = 5.0],  
4   [OrderID = 3, CustomerID = 2, Item = "fishing net", Price = 25.0]}),  
5   #"Capitalized Each Word" = Table.TransformColumns(Orders, {"Item", Text.Proper})  
6 in  
7   #"Capitalized Each Word"
```

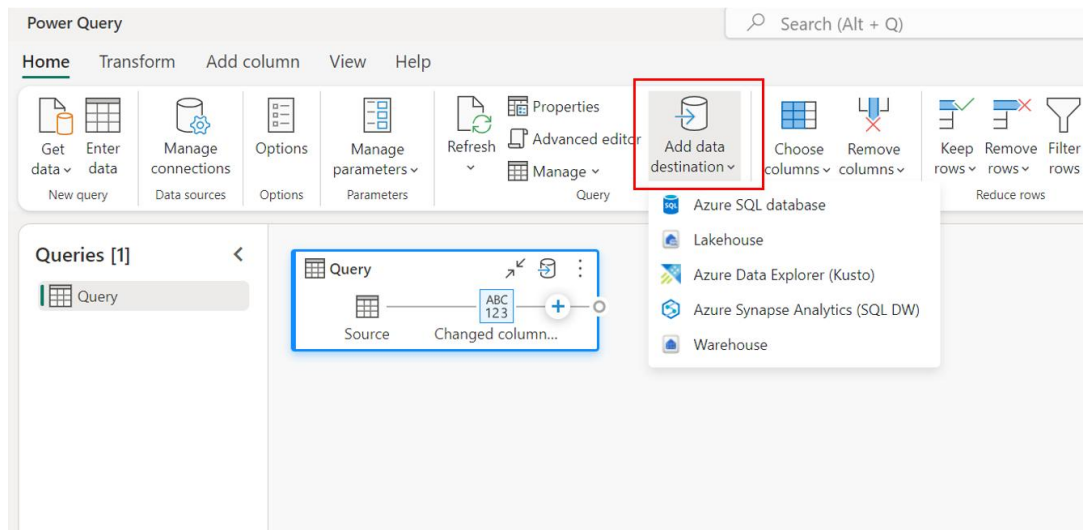
OK

Cancel

Everywhere

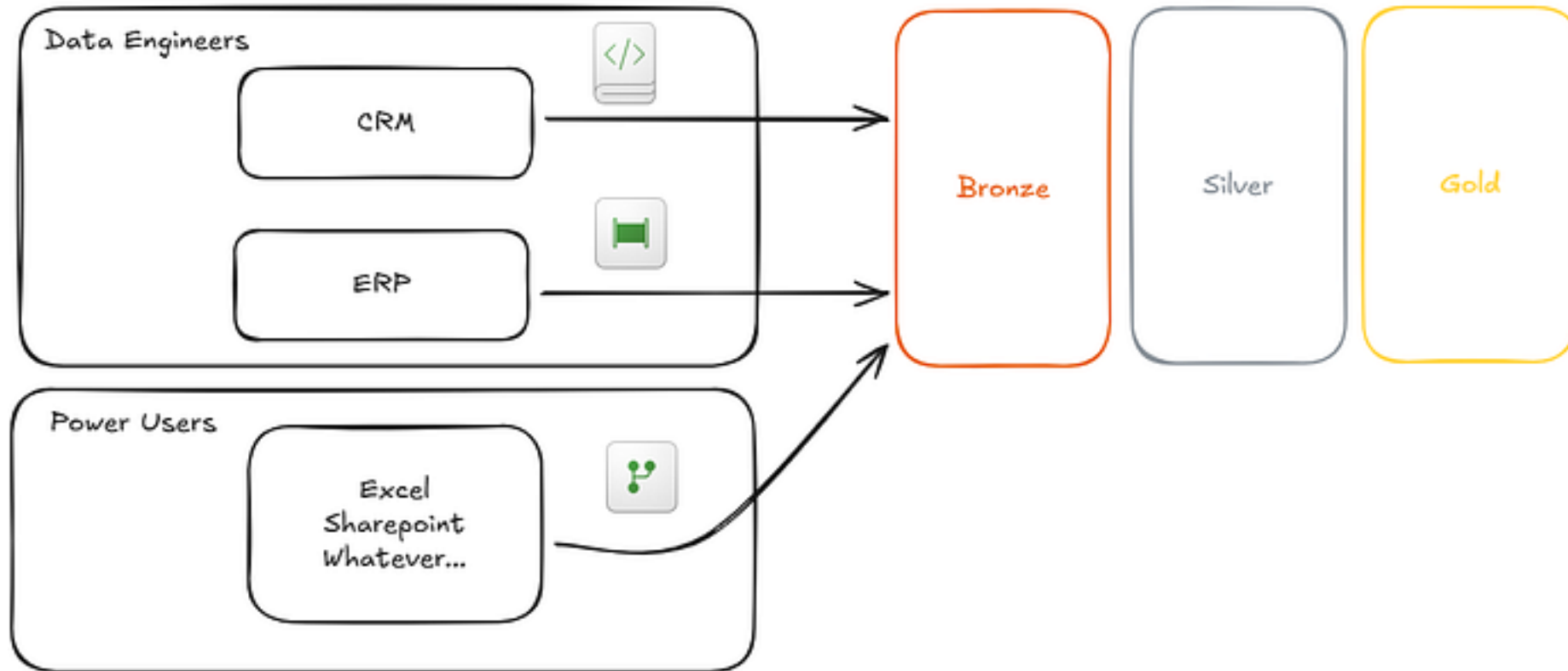
Product	M engine ¹	Power Query Desktop ²	Power Query Online ³	Dataflows ⁴
Excel for Windows	Yes	Yes	No	No
Excel for Mac	Yes	Yes	No	No
Power BI	Yes	Yes	Yes	Yes
Power Apps	Yes	No	Yes	Yes
Power Automate	Yes	No	Yes	No
Power BI Report Server	Yes	Yes	No	No
Azure Data Factory	Yes	No	Yes	Yes
Data Factory in Microsoft Fabric	Yes	No	Yes	Yes
SQL Server Integration Services	Yes	No	No	No
SQL Server Analysis Services	Yes	Yes	No	No
Dynamics 365 Customer Insights	Yes	No	Yes	Yes

Read and... Write!



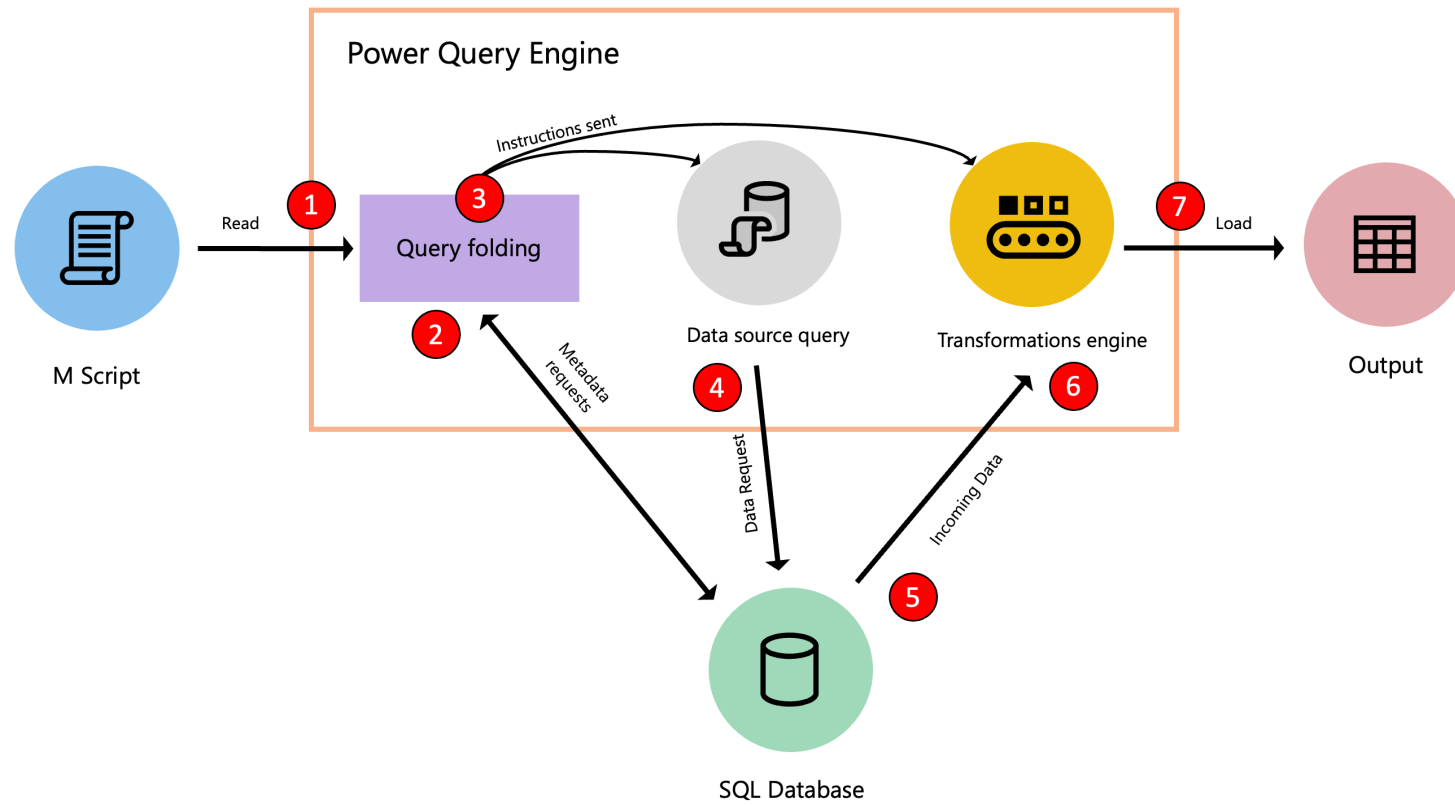
- Azure SQL databases
- Azure Data Explorer (Kusto)
- Azure Datalake Gen2
- Fabric Lakehouse Tables
- Fabric Lakehouse Files*
- Fabric Warehouse
- Fabric KQL database
- Fabric SQL database
- SharePoint Files*
- Snowflake database
- *CSV and Excel

Power Query is ETL democratization



[ETL democratization with Dataflows Gen2 | by Riccardo Perico | Rik in a Data Journey | Medium](#)

Query folding





*“Data should be transformed as far upstream as possible,
and as far downstream as necessary.”*

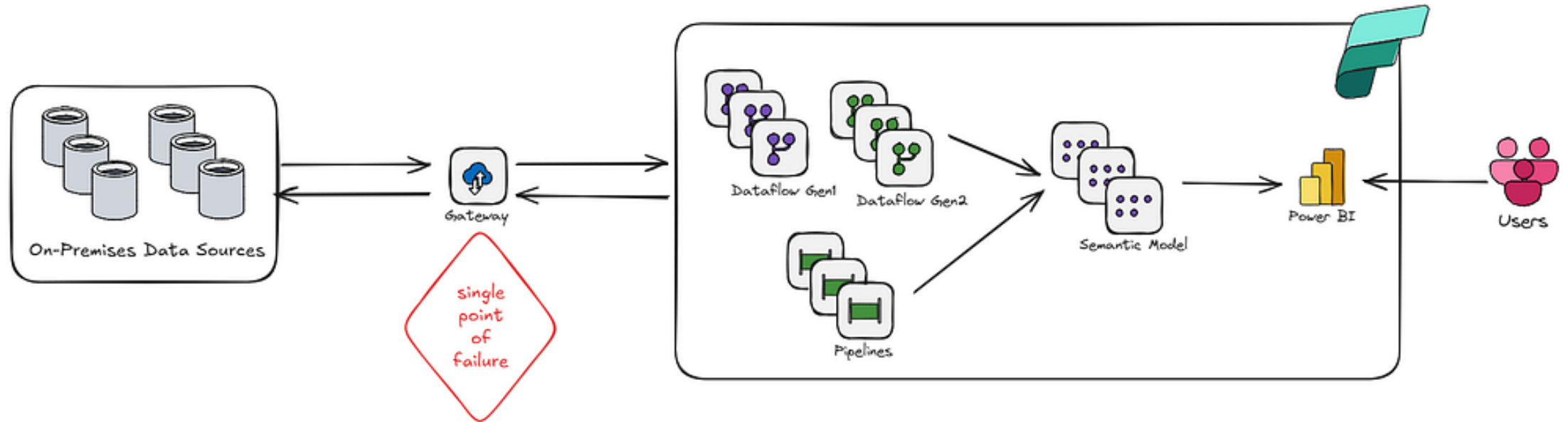
Matthew Roche

Principal Program Manager: Fabric CAT at Microsoft



On-Prem Gateway

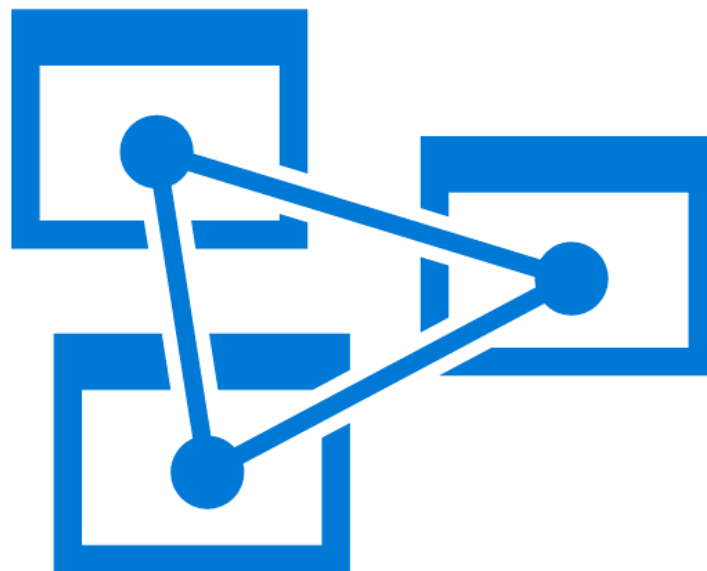
More important than you think





Modeling

Again... behind the scenes



It's not for small models only

SKU	Limit ¹
F2	1 GB
F4	2 GB
F8/EM1/A1	3 GB
F16/EM2/A2	5 GB
F32/EM3/A3	6 GB
F64/P1/A4	10 GB
F128/P2/A5	10 GB
F256/P3/A6	10 GB
F512/P4/A7	10 GB
F1024/P5/A8	10 GB
F2048	10 GB

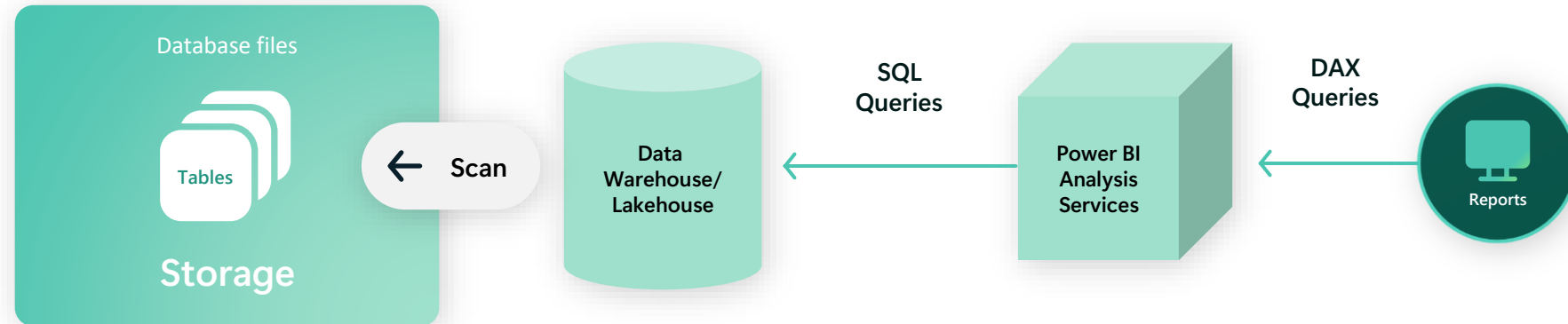


* Up to available memory on your capacity, excluding memory required for query execution and processing, which is often 20-30% of total memory.

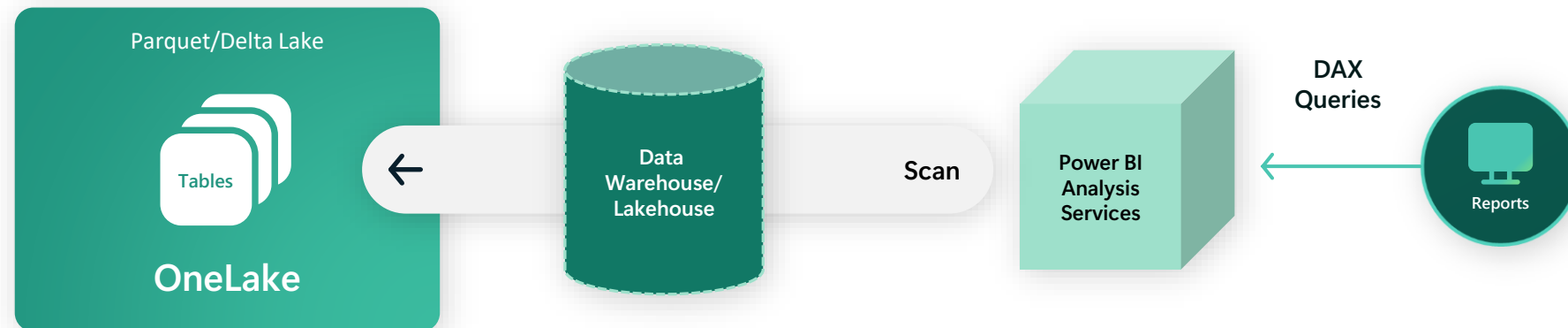
S9V2	Standard	★
1,280	Query Processing Units	
	Up to 400 GB Cache	
	Dedicated service	
	SSL	
		20,629.41
		EUR/MONTH (ESTIMATED)

400GB is not enough?

Direct Query Mode



Direct Lake Mode



Modeling is a real thing

- Modeling is the most important stuff in Power BI project
- Good Star Schema =
 - Great Performance
 - As much data as needed
 - Easy Dax
 - Everyone's happiness

Vertipaq under the hood

- Columnar engine
- Encoding for compression
 - Value Encoding
 - Hash Encoding
 - RLE (Run-length-encoding)

Atomic bomb in a column

1 parquet

1 column

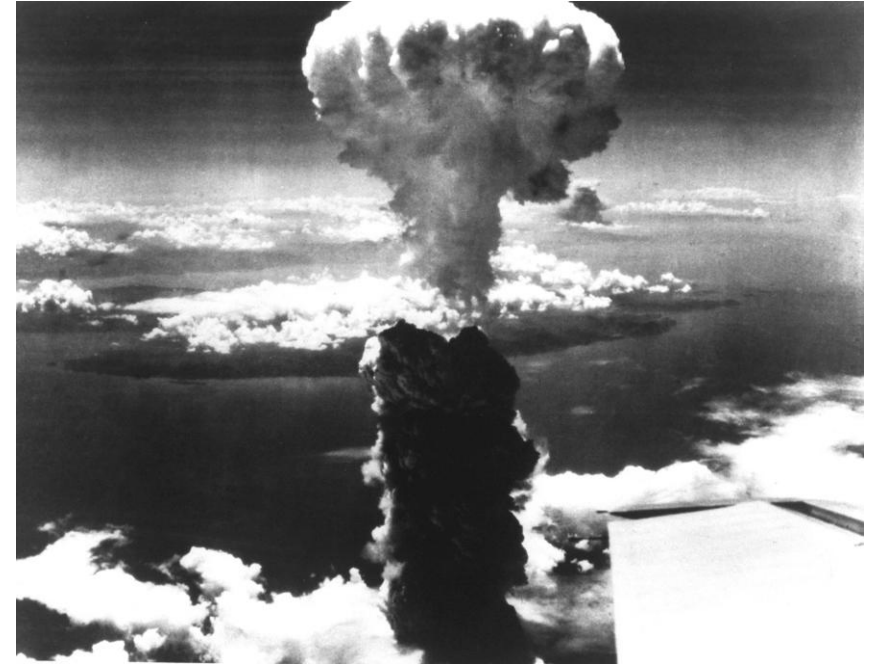
622 trillion “zeros”

... on file system

42 KB

... in memory

4 PB



[42.parquet – A Zip Bomb for the Big Data Age – DuckDB](#)



Always tie to best
practices



Secure by default

Don't duplicate that report, use RLS!

Manage security roles

Create new security roles and use filters to define row-level data restrictions.

Roles

+ New

Untitled ...

Tables

- Customer ...
- Date ...
- Product ...
- Reseller ...
- Sales ...
- Sales Order ...
- Sales Territory ...

Rules

+ New ☒ Select all Delete Group Ungroup

Show data if All of these rules are true

	Column	Condition	Value
☐	Region	Equals	West
☐	Country	Equals	United States
☐	Group	Equals	A

+ New

Switch to DAX editor

Row-Level Security

Eastern US (0)

Members (0)

People or groups who belong to this role

Enter email addresses

Add

Save

Close

You can hide even tables and columns

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Table Permissions' folder is expanded, showing a list of tables including 'Categories', 'Customers', and several 'LocalDateTable' entries. The 'Categories' table is selected, and its permissions are shown in the right pane. The 'Permissions' column for 'Categories' is set to 'None', which is highlighted with a red box. Below the table list, the 'Translations, Perspectives, Security' folder is also expanded, showing 'Members' and 'Model Permission'.

In the bottom right pane, a query is being executed. The query text is as follows:

```
//Run as OLS Test  
EVALUATE  
SELECTCOLUMNS (  
    DimCustomer,  
    "Last Name", DimCustomer[LastName],  
    "Birth Date", DimCustomer[BirthDate]  
)
```

The query execution results pane at the bottom shows an error message, which is highlighted with a red box:

```
Query (6, 19) Column 'BirthDate' in table 'DimCustomer' cannot be found or may not be used in this expression.  
Run complete
```



Reporting

File Home Insert Modeling View Optimize Help External tools

Paste Copy Cut Get data Excel OneLake SQL Enter data Dataverse Recent sources Transform Refresh data Queries New visual Text box More visuals New visual calculation measure measure Sensitivity Publish Prep data for Copilot AI

Microsoft | Skateboard Store Last Refresh: Sep 30, 2025 / Chicago, IL, USA

OneNote Office 365

Return Rate 24%

What If... We Decrease Our Return Rate (%) to: 25

Net Sales (Forecast) \$30,772

Extra Profit \$0

Net Sales vs "What If" Analysis

Store	Net Sales	Forecast	Extra Profit
Abbas	\$2,646	\$2,646	\$0
Aliqui	\$3,724	\$3,724	\$0
Barba	\$1,568	\$1,568	\$0
Contoso	\$2,842	\$2,842	\$0
Fama	\$3,626	\$3,626	\$0
Leo	\$1,764	\$1,764	\$0
Natura	\$1,274	\$1,274	\$0
Palma	\$0	\$0	\$0
Pirum	\$2,744	\$2,744	\$0
Pomum	\$1,372	\$1,372	\$0
Quibus	\$2,940	\$2,940	\$0
Salvus	\$2,646	\$2,646	\$0
VanArsdel	\$2,548	\$2,548	\$0
Victoria	\$1,078	\$1,078	\$0
Total	\$30,772	\$30,772	\$0

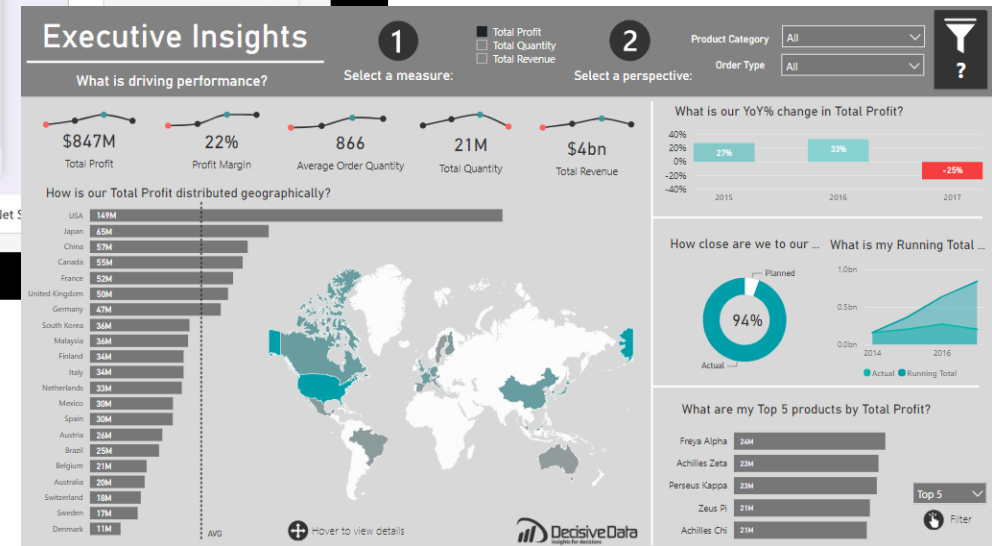
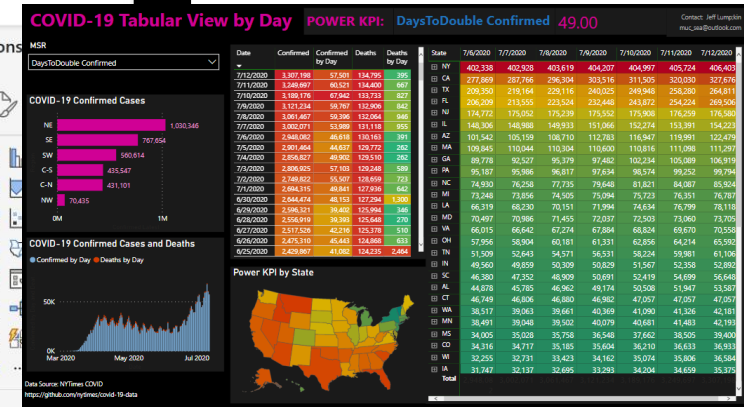
"What If" Analysis Forecast

Predicted Net Sales Forecast

Return Rate 24%

Feb 2019 Mar 2019 Apr 2019 May 2019 Jun 2019

Page 10 of 11



Microsoft Teams

Search or type a command

General Posts Files Wiki Customer 360 +

Teams

- Your teams
- Game Marketing Org
- General
- Complete
- Launches
- Gaming Insights Support Team
- General
- Support Questions

Contoso

Top 10 Customers in Revenue

Customer	Revenue	Estimates	Variance
Proseware, Inc.	45,269,173	45,143,310	-125,863
Litware	24,885,124	40,919,281	-16,034,157
Humongous Insurance	21,046,152	27,963,310	-6,917,158
Wide World Imports	17,859,351	24,421,193	-6,561,842
Blue Yonder Airlines	15,819,171	23,072,719	-7,253,548
Wingtip Toys	13,762,010	14,498,560	-736,550
The Phone Company	12,620,120	24,714,620	-12,094,500
Total	263,678,272	426,898,683	-163,220,411

Revenue by Location

Revenue by Status

SAR (19,72%)

Don Open Post

Revenue by Closed Date

Open Opportunity by Sales Stage

Qualify Propose Develop Close

View by Rate View by Rank

Status	Revenue	Revenue %
Won	218,678,025	78.67%
Lost	219,876,525	78.67%
Open	53,665,504	19.20%
In Progress	50,797,453	18.17%
Delayed	2,886,521	1.03%
Lost	5,863,143	2.13%
Can Sell	6,000,000	1.42%
Canceled	1,965,146	0.70%
Total	279,187,692	100.00%

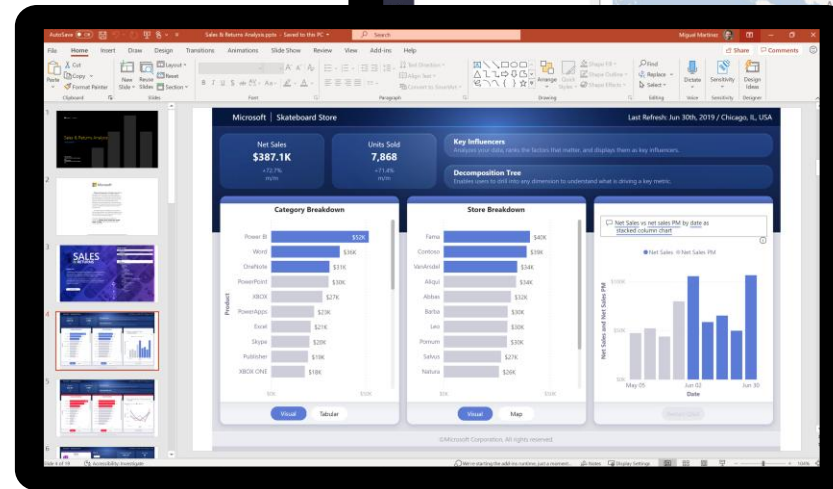
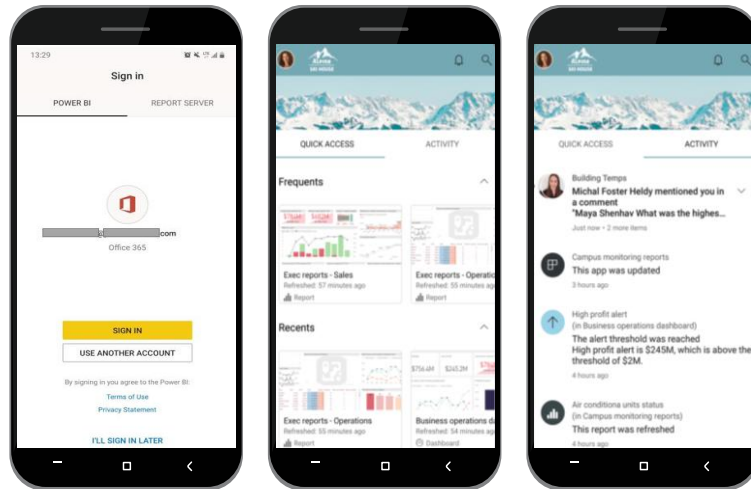
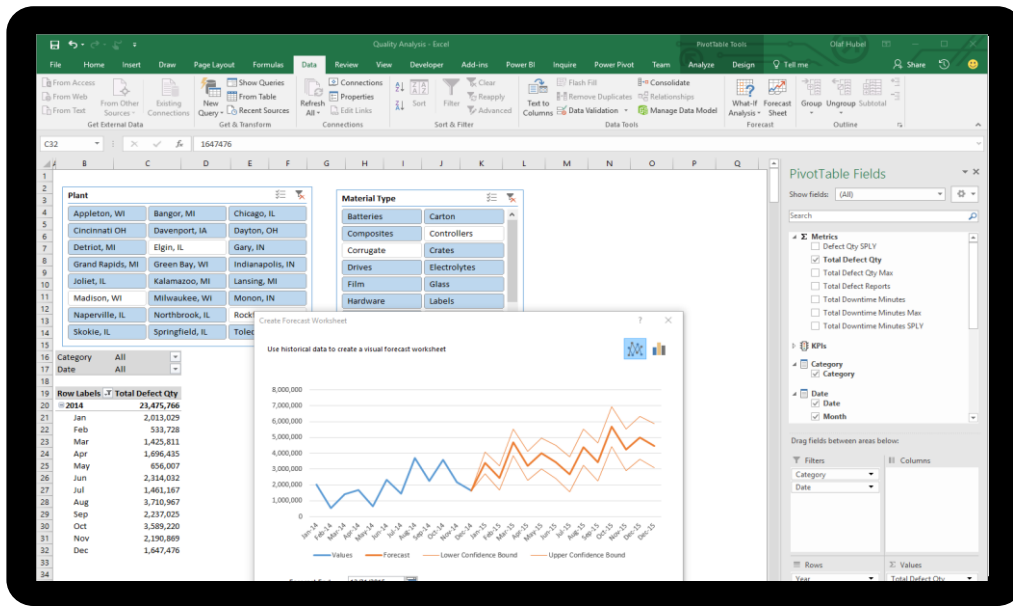
Revenue by Product - Rank

Product	Rank
Litware Laptop Keyboard K103	3.2M
Litware Wireless Mouse M103	3.2M
Proseware All-in-One Printer P103	3.2M
Southridge S17 LCD	3.2M
WWP Desktop PC C33 G230	3.2M
9609	2.9M
Egabrium Gaming H16 M6000	2.3M
Fabrikam Tablet T4100	1.3M
3D Printer	0.2M

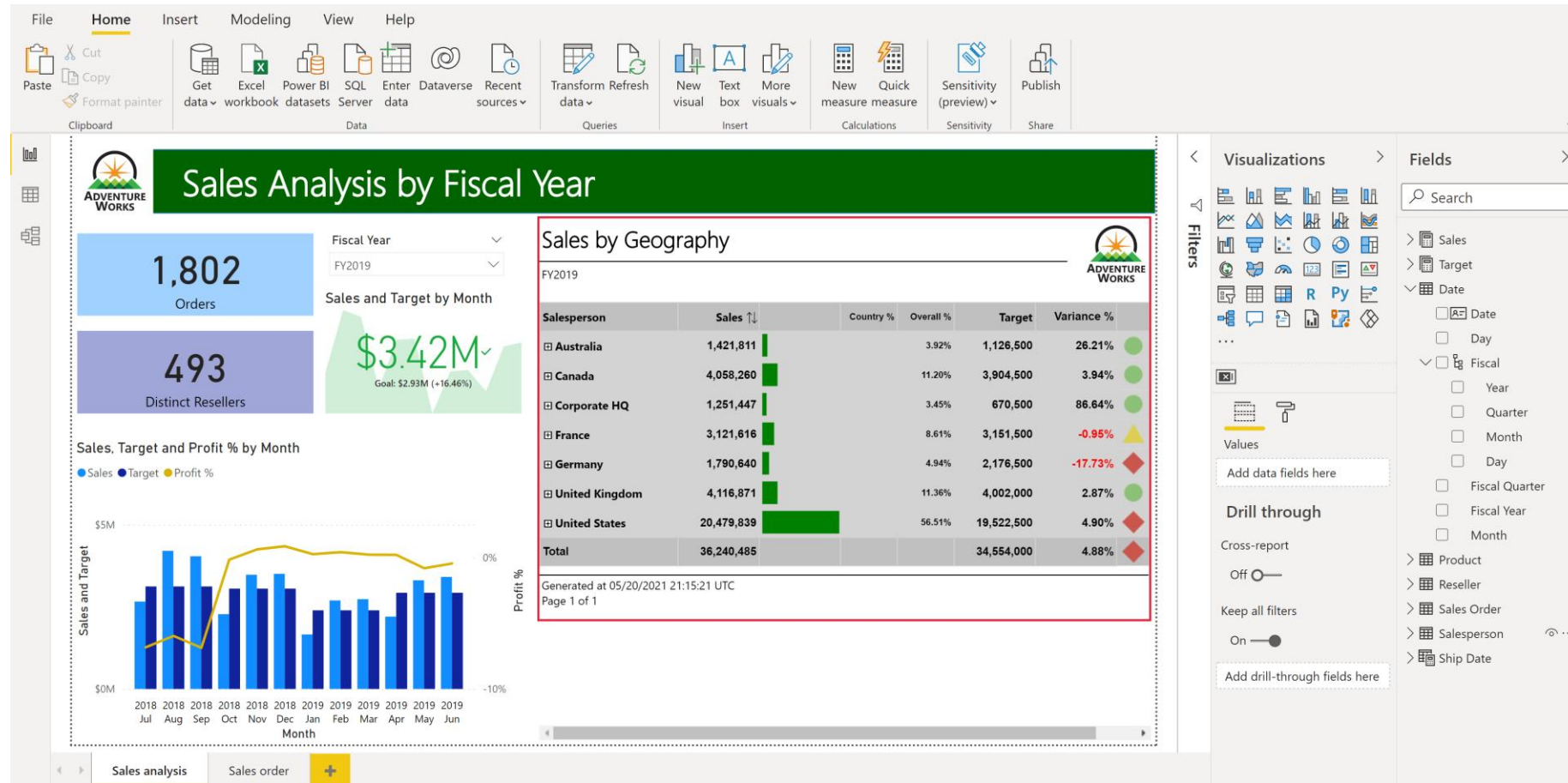
Case Count by Status

Status	Count
In Progress	158
Problem Solved	11
Cancelled	1

Summary: Total Case Count: 158, Case Resolved %: 7.0%



Long live paginated reports





DAX

Modeling and Querying language

- Modeling

- Measures
- Calculated Columns
- Calculated Tables

→ Total Sales = SUM([Sales Amount])

→ = [Calendar Year] & " Q" & [Calendar Quarter]

→

```
=SUMMARIZECOLUMNS(  
    DimProduct[Color] , DimCurrency[CurrencyName]  
    , "Sales" , SUM(FactInternetSales[SalesAmount])  
    )
```

- Row-level security

→ = Customers[Country] = "USA"

- Querying

```
EVALUATE ( FILTER ( 'DimProduct', [SafetyStockLevel] < 200 ) )  
ORDER BY [EnglishProductName] ASC
```

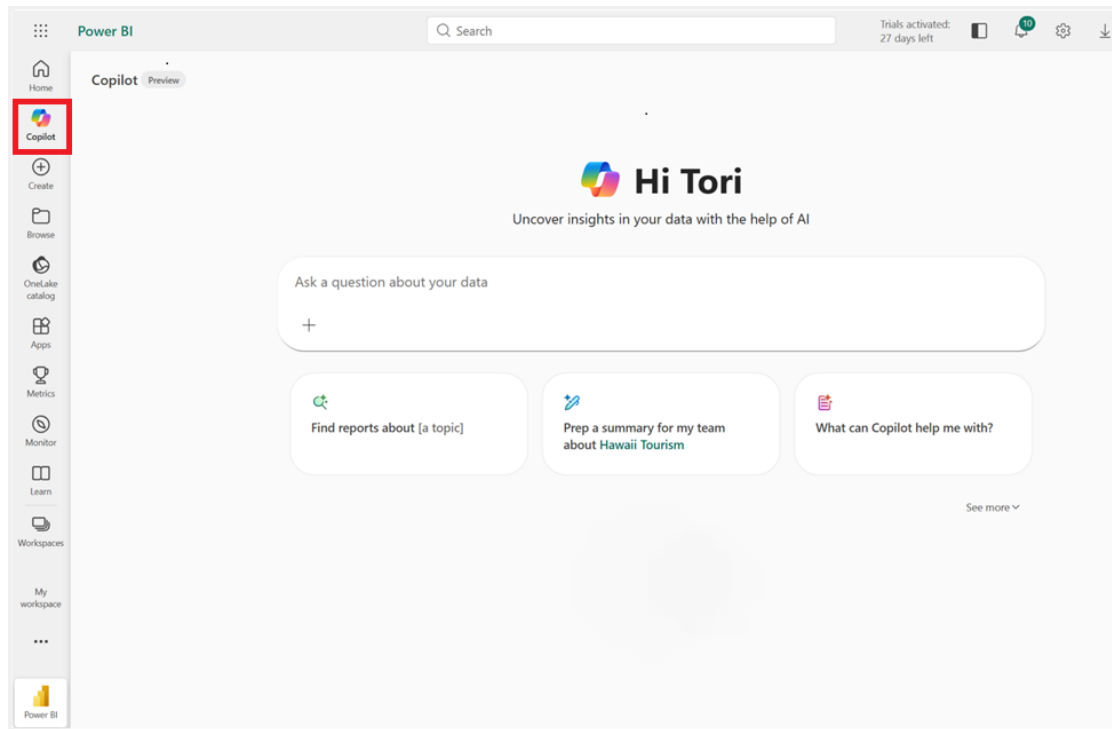


Solve in model, not in DAX!
You're not Marco Russo 😊



Ok... let's talk about AI

Your Power BI chatbot



Find items about sales revenue

Copilot AI-generated content may be incorrect

Sure, I found the following items:

- [Contoso Sales](#)
Workspace: Contoso Green - Stores a...
company's revenue performance, sales forecast accuracy, closing percentages, and revenue data by territory and product
This report was matched on the title and the description.
Within app: [Business Overview](#)
- [Contoso Sales](#)
Workspace: PBIFabricDoNotDelete
This model was matched on the title and its contents. It also contains a verified answer similar to your query.
- [AdventureWorks](#)
Workspace: Contoso Green - Stores a...
sales performance of different product categories, highlights trends, and evaluates the impact of discounts on sales
This report was matched on the description.
Within app: [Business Overview](#)
- [AdventureWorks Sales \(2\)](#)
Workspace: CopilotTesting | **Opened:** 8 months ago
This report was matched on the title and its contents.

Chat with your data

customer-sales How many transactions were there for each product?

 **Copilot** AI-generated content may be incorrect

Here are the number of transactions for each product:

- **Credit Card:** 10 transactions
- **Savings Account:** 6 transactions
- **Checking Account:** 2 transactions
- **Car Insurance:** 5 transactions

1  customer-sales

👍 🗑️

What is the most common product purchased by customers?

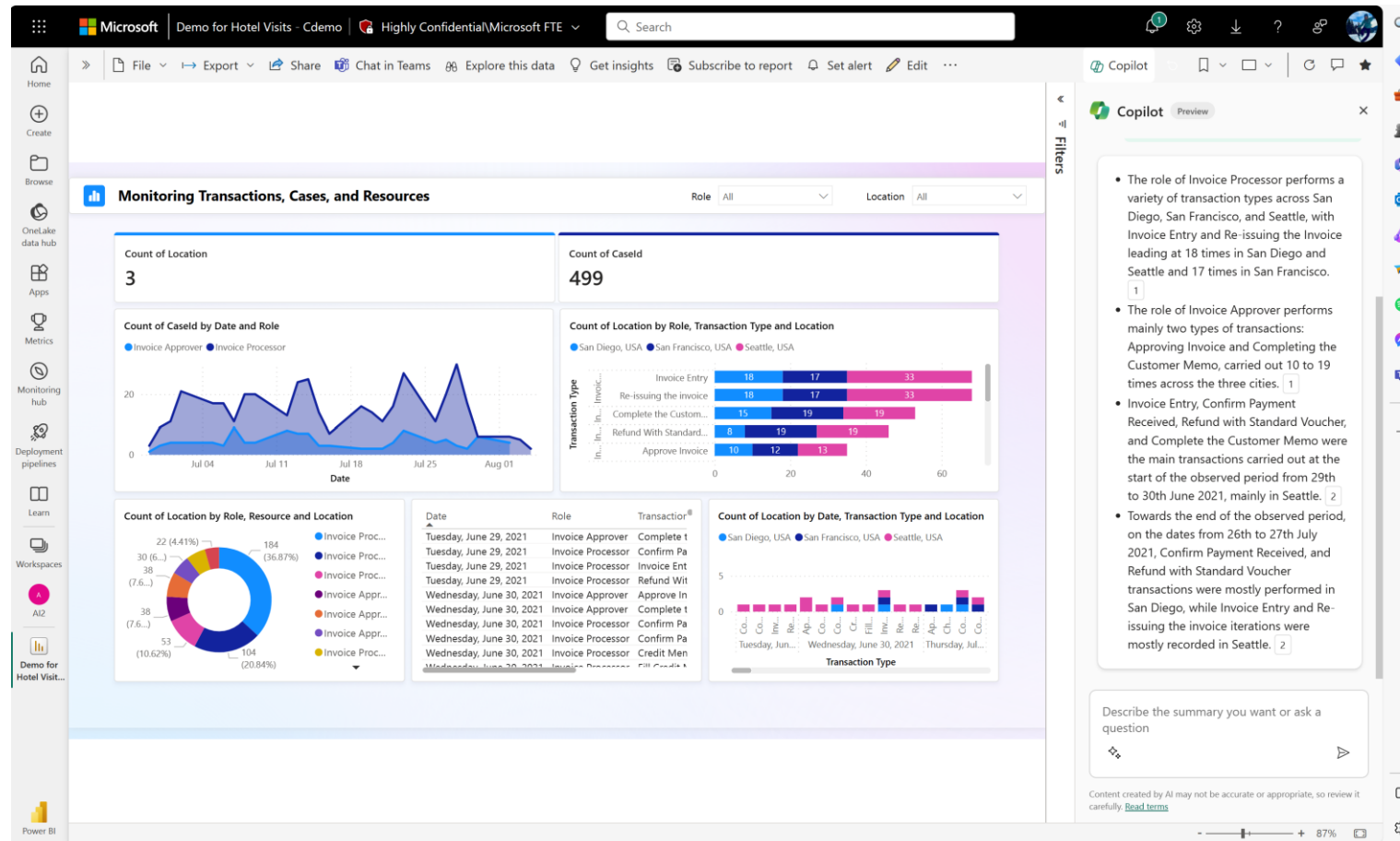
 **Copilot** AI-generated content may be incorrect

The most common product purchased by customers is the **Credit Card**, with a total of 10 transactions.

1  customer-sales

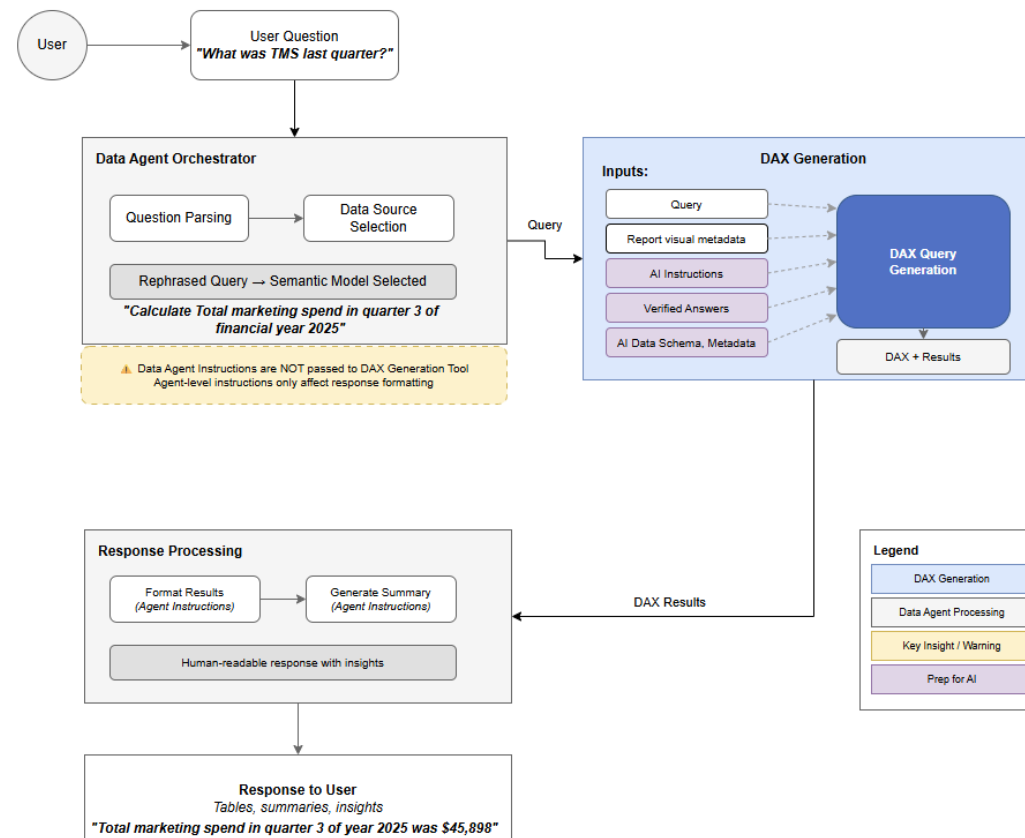
👍 🗑️

Let it summarize the report



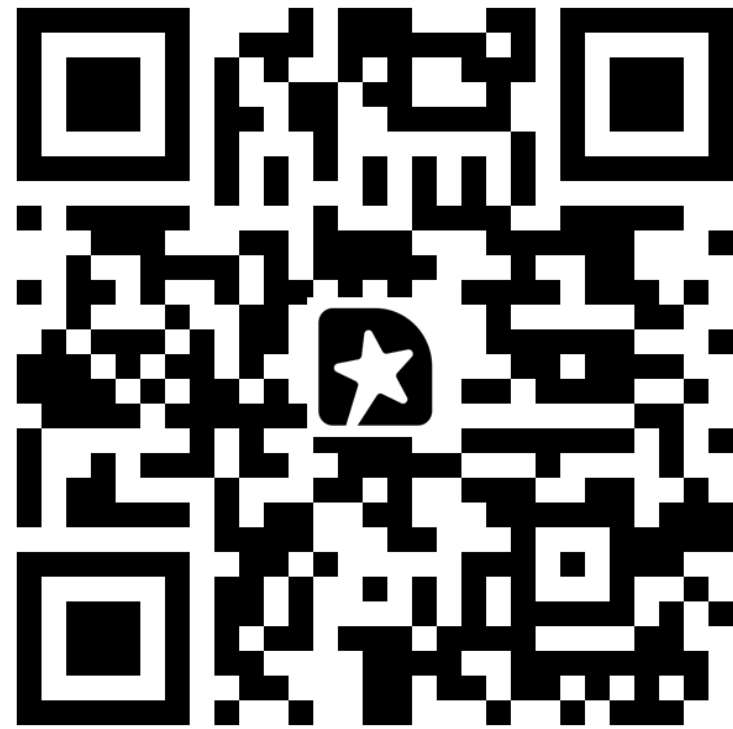
Consumed in Fabric Data Agents

Fabric Data Agent: Query Processing Flow for Semantic Models





Q&A



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 medium.com/riccardo-perico

BI Architect @ **Lucient**

+15 years in Microsoft “Data Realm”

Giving my little contribution to the community





Data, Models, and Magic: Getting Started with Power BI

Riccardo Perico

BI Architect @ Lucient Italia | Data Platform MVP